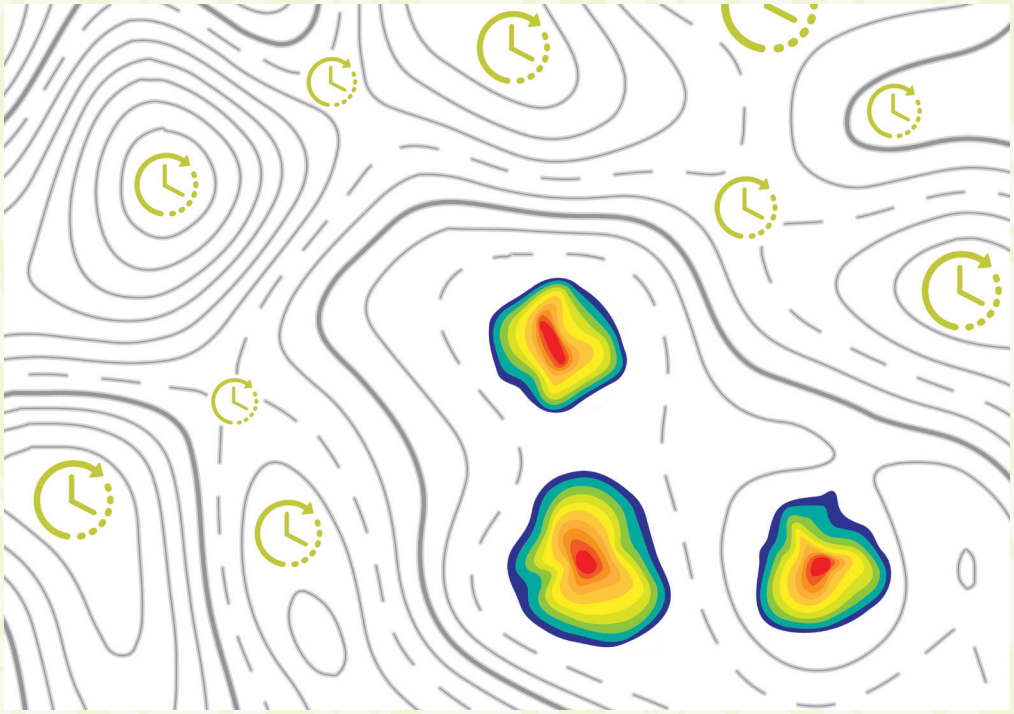
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Toward Environmentally Equitable AI

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THE GROWING ADOPTION of artificial intelligence (AI) has been accelerating across all parts of society, boosting productivity and addressing pressing global challenges such as climate change. Nonetheless, the technological advancement of AI relies on computationally intensive calculations and thus has led to a surge in resource usage and energy consumption. Even putting aside the environmental toll of server manufacturing and supply chains, AI systems can create a huge environmental cost to communities and regions where they are deployed, including air/thermal pollution due to fossil fuel-based electricity generation and further-stressed water resources due to AI's staggering water footprint.[12,25] To make AI more environmentally friendly and ensure its overall impacts on climate change are positive, recent studies have pursued multifaceted approaches, including efficient training and inference,[5] energy-efficient GPU and accelerator designs,[19] carbon forecasting,[14] carbon-aware task scheduling,[1,21] green cloud infrastructures,[2] sustainable AI policies,[10,18] and more. Additionally, datacenter operators have also increasingly adopted carbon-free energy (such as solar and wind power) and climate-conscious cooling systems, lowering carbon footprint and direct water consumption.[8]

Minimizing the total environmental cost of a globally deployed AI system across multiple regions does not necessarily mean each region is treated equitably.

Although these initiatives are encouraging, unfortunately, a worrisome outcome—*environmental inequity*—has emerged.[3] That is, minimizing the total environmental cost of a globally deployed AI system across multiple regions does not necessarily mean each region is treated equitably. In fact, the environmental cost of AI is often disproportionately higher in certain regions than in others. Even worse, AI's environmental inequity can be amplified by existing environmental equity-*agnostic* resource allocation, load balancing, and scheduling algorithms—and compounded by enduring socioeconomic disparities between regions. For example, geographical load-balancing (GLB) algorithms that aggressively exploit regional differences to seek lower electricity prices and/or more renewables[7,17] may schedule more workloads to water-inefficient datacenters (located in, for example, water-stressed Arizona), resulting in a disproportionately high water footprint and adding further pressures to local water supplies.[9]

Addressing the emerging environmental inequity is becoming an

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integral part of responsible AI.[3] It has increasingly received public attention and urgent calls for mitigation efforts. For example, the AI Now Institute compares the uneven regional distribution of AI's environmental costs to "historical practices of settler colonialism and racial capitalism" in its 2023 Landscape report.[11] The United Nations Educational, Scientific and Cultural Organization (UNESCO) recommends against the usage of AI if it creates "disproportionate negative impacts on the environment."[23] California recognizes the need for "ensuring environmental costs are equitably distributed" in its State Report.[4] Environmental justice is ranked by Meta as the most critical factor among all environmental-related topics.[15]

In this article, we advocate *environmental equity* as a priority for the management of future globally deployed AI systems. Concretely, we explore the potential of harnessing AI workloads' scheduling flexibility and using equity-aware GLB as a lever to fairly redistribute the environmental cost across regions, ensuring that no single region disproportionately bears the environmental burden. Then, we present key algorithmic challenges to enable AI's environmental equity without significantly degrading the other performance metrics, such as energy cost and inference accuracy. Finally, we discuss future directions to unleash the full potential of system management for environmentally equitable AI, including coordinated scheduling of AI training and inference, joint optimization of IT and non-IT resources, holistic control of system knobs, and building theoretical foundations.

Our proposal of environmental equity advances the boundaries of existing research on sustainable AI and mitigates the otherwise uneven distribution of AI's environmental costs across different regions. Equity and fairness are also crucial considerations for AI. The existing research in this space has predominantly tackled prediction unfairness against certain individuals and/or groups.[20,26] Thus, environmental equity adds a unique dimension of fairness and significantly complements the existing literature, collaboratively building equitable and responsible AI.

Opportunities and Challenges for Equity-Aware GLB

In this section, we present the potential opportunities of leveraging equity-aware GLB to fairly redistribute the environmental cost across different regions, followed by algorithmic challenges.

Opportunities. The limited power-grid capacity has necessitated increasing flexibility from datacenters to support demand response and maintain grid stability. A notable example is the recent industry initiative to maximize load flexibility for grid-integrated datacenters.[6] Specifically, AI workloads exhibit three primary types of flexibility: *spatial*, where AI training and inference tasks can be distributed across multiple datacenters with minimal impact on latency; *temporal*, in which AI training tasks can be executed intermittently, provided they meet a given deadline; and *performance*, where a single inference request can be processed by different AI models, each offering distinct trade-offs between accuracy and resource consumption.

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These flexibilities can be exploited to promote environmental equity while satisfying other performance objectives. To achieve this, we can leverage a variety of approaches, such as AI computing resource allocation, load balancing, and job scheduling, which we collectively refer to as system *knobs*.

In practice, the datacenter fleet of large companies, such as Google and Microsoft, often includes a few tens of self-managed hyperscale datacenters and many more leased third-party colocation datacenter spaces spreading throughout the world.[8] By renting virtual machines on public clouds, even a small business can flexibly choose its deployment region and place its computing workloads accordingly. As such, GLB is an important and common knob that can spatially balance computing workloads' energy demand as well as environmental footprint across different locations.

As a concrete example, we consider moving AI inference workloads around from one datacenter to another and exploit equity-aware GLB to mitigate AI's environmental inequity. To achieve equitable distribution of AI's environmental cost, we consider the notion of *minimax* fairness. Mathematically, denoting $x_{i,t}$ as the amount of AI workloads processed in datacenter i at time t and $E_{i,t}\left(x_{i,t}\right)$ as the resulting regional environmental cost (for example, due to water consumption[12] and air/thermal/waste pollution from non-renewable energy[24]), we consider an equity-aware objective: $\sum_{t=1}^{T}\sum_{i} cost_{i,t}\left(x_{i,t}\right) + \lambda \cdot \max_{i}\left[\sum_{t=1}^{T} E_{i,t}\left(x_{i,t}\right)\right]$, where the first term is the traditional GLB cost (for example, total carbon/water footprint and energy cost) specified based on the prior literature,[9] the second term "$\max_{i}\left[\sum_{t=1}^{T} E_{i,t}\left(x_{i,t}\right)\right]$" serves as the equity regularizer by reducing the highest regional environmental cost, and $\lambda \geq 0$ is the weight.

A snapshot of results. We run a simulation based on the BLOOM model (a large language model) inference trace deployed in 10 different datacenters throughout the world and show a snapshot of our results in

Table 1. The details of the simulation are available in Li et al.[13] We consider both *full* GLB (that is, each request can be flexibly routed to any datacenter) and *partial* GLB (that is, each request can only be routed to a subset of datacenters depending on its originating location). Compared to common baseline algorithms that simply minimize the total energy cost (`GLB-Cost`), carbon emission (`GLB-Carbon`), or workload-to-datacenter distance (`GLB-Dist`), our algorithm (called `eGLB-Off`) can effectively mitigate the environmental inequity by reducing the ratio of the maximum to the average regional environmental footprint. Importantly, while there is an inevitable conflict between minimizing the total cost/environmental footprint and addressing the environmental inequity, `eGLB-Off` can still keep the total cost reasonably low. Additionally, we study a simple online algorithm (called `eGLB`) based on dual mirror descent to show the potential of mitigating environmental inequity in an online setting. While there is a gap between `eGLB` and `eGLB-Off` due to online informational constraints, `eGLB` outperforms the equity-unaware baseline algorithms in terms of the environmental footprint's peak-to-average ratio, demonstrating the potential of online GLB to mitigate AI's environmental inequity.

Table 1. Comparison of GLB algorithms in terms of the total energy cost and the normalized water/carbon peak-to-average ratio (PAR). Details in Li et al.[13]

		Algorithm				
GLB	**Metric**	`GLB-Cost`	`GLB-Carbon`	`GLB-Dist`	`eGLB-Off`	**eGLB**
Full	**Cost** (US$)	29170	45535	47038	33669	**33752**
	PAR (Water)	1.71	1.85	1.44	1.27	**1.37**
	PAR (Carbon)	1.68	1.70	1.41	1.13	**1.22**
Partial	**Cost** (US$)	29659	45535	47038	34186	**34162**
	PAR (Water)	1.72	1.84	1.44	1.30	**1.38**
	PAR (Carbon)	1.69	1.71	1.41	1.12	**1.22**

Challenges. While equity-aware GLB can potentially mitigate AI's environmental inequity, the equity regularizer "$\max_i \left[\sum_{t=1}^{T} E_{i,t}\left(x_{i,t}\right)\right]$" fundamentally separates our problem from the existing sustainable GLB approaches and creates substantial algorithmic challenges. Specifically, the equity cost "$\max_i \left[\sum_{t=1}^{T} E_{i,t}\left(x_{i,t}\right)\right]$" is unknown until the end of T time slots, but complete future information (for example, future workload arrivals and water efficiency) may not be perfectly known in advance. Moreover, even though prediction is often available in practice, it may not be accurate, and its untrusted nature means we cannot simply take the prediction as if it were the ground truth.

Additionally, the traditional design of online competitive algorithms often focuses on guaranteeing the worst-case performance robustness. But, the resulting average performance can be far from optimal due to the conservativeness needed to address potentially worst instances. By contrast, machine learning (ML)-based optimizers (for example, reinforcement learning) can improve the average performance of online decision making by exploiting rich historical data and statistical information, but they typically sacrifice the strong performance robustness needed by real AI systems, especially when there is a distributional shift, the ML model capacity is limited, and/or inputs are adversarial. Thus, to achieve the best of both worlds while pursuing online equitable-aware GLB, we have to carefully balance the usage of traditional competitive algorithms and ML-based optimizers by designing new learning-augmented online algorithms.

Tackling the challenges raised by environmental inequity in modern planet-scale AI systems requires a revisit to the algorithmic foundations and the development of new theoretical tools.

Future Directions

We discuss a few future directions to leverage system knobs for environmentally equitable AI.

Coordinated scheduling of AI training and inference. While AI inference offers spatial flexibility, AI model training has great *temporal* scheduling flexibility, as we can choose *when* to train the AI models in a stop-and-go manner. We can also choose where to perform AI model training and even possibly change the locations in the middle of the training process. Thus, a potential direction is to explore coordinated scheduling of AI training and inference tasks to fairly distribute AI's overall environmental costs across different regions.

Joint optimization of IT and non-IT resources. Datacenters have increasingly begun to install onsite carbon-free energy, such as solar power, to partially power the workloads and lower the environmental footprint.[21] However, renewables are often intermittent, and the available energy-storage capacity is finite. Thus, how to optimize AI demand response given intermittent renewables is challenging, yet worth investigating for addressing AI's environmental inequity.

Holistic control of system knobs. In addition to GLB, a rich set of system knobs are available and of-

fer flexible trade-offs, such as dynamic model selection for inference, turning servers on/off, and resource allocation to different AI tasks. For example, different AI models can exhibit different energy-accuracy trade-offs for the same task. Holistic control of these system knobs holds enormous potential to curb AI's resource usage and mitigate environmental inequity, but also presents additional challenges due to the significantly enlarged decision space.

Theoretical foundations. Optimizing a variety of system knobs for environmentally equitable AI has its roots in *fair* decision making, a classical area that bridges computer systems and algorithms and enjoys a long history with rich theoretical results[16,22] and prominent production deployments. However, this classic literature primarily focuses on algorithms that ensure that different job or flow types receive a fair share of system resources, including CPU and memory. Tackling the challenges raised by environmental inequity in modern planet-scale AI systems requires a revisit to the algorithmic foundations and the development of new theoretical tools, which can systematically capture the conflicts between traditional measures of performance, such as accuracy and latency, with measures of emerging importance, such as environmental equity. Thus, it is crucial to build new theoretical foundations to support the design of environmentally equitable AI.

Conclusion

In light of AI's wildly different environmental costs across different regions, we advocate *environmental equity* as a priority for the management of future AI systems. We present the potential opportunities and algorithmic challenges of tapping into AI workloads' scheduling flexibility and leveraging equity-aware GLB to mitigate AI's environmental inequity. Finally, we discuss a few future directions to unleash the full potential of system knobs for environmentally equitable AI, including coordinated scheduling of AI training and inference, joint optimization of IT and non-IT resources, holistic control of system knobs, and building theoretical foundations. Our proposal of environmental equity pushes forward the boundaries of existing system management for sustainable AI and also adds a unique dimension to AI fairness, collaboratively building equitable and responsible AI.

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In light of AI's wildly different environmental costs across different regions, we advocate environmental equity as a priority for the management of future AI systems.

References

1. Acun, B. et al. Carbon explorer: A holistic framework for designing carbon aware datacenters. In *Proceedings of the 28th ACM Intern. Conf. on Architectural Support for Programming Languages and Operating Systems 2*. Association for Computing Machinery (2023), 118–132.
2. Bashir, N. et al. Enabling sustainable clouds: The case for virtualizing the energy system. In *Proceedings of the ACM Symp. on Cloud Computing*. Association for Computing Machinery (2021), 350–358.
3. Bender, E.M., Gebru, T., McMillan-Major, A., and Shmitchell, S. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conf. on Fairness, Accountability, and Transparency*. Association for Computing Machinery, 610–623.
4. California Government Operations Agency. Benefits and risks of generative artificial intelligence report. *State of California Report* (Nov. 2023).
5. Chen, L., Zaharia, M., and Zou, J. FrugalGPT: How to use large language models while reducing cost and improving performance. (2023); arXiv:2305.05176 [cs.LG]
6. EPRI. DCFlex Initiative. (2024); https://msites.epri.com/dcflex.
7. Gao, P.X., Curtis, A.R., Wong, B., and Keshav, S. It's not easy being green. *SIGCOMM Comput. Commun. Rev.* (2012).
8. Google. Environmental Report. (2023); https://sustainability.google/reports/.
9. Islam, M.A. et al. Exploiting spatio-temporal diversity for water saving in geo-distributed data centers. *IEEE Transactions on Cloud Computing 6*, 3 (2018), 734–746.
10. ISO/IEC JTC for AI (SC42). ISO/IEC TR 20226 Sustainability: Harnessing the power of AI. (2023); https://tinyurl.com/2bj6m6dz.
11. Kak, A. and Myers, S. AI Now 2023 landscape: Confronting tech power. *AI Now Institute* (April 2023).
12. Li, P., Yang, J., Islam, M.A., and Ren, S. Making AI less "thirsty": Uncovering and addressing the secret water footprint of AI models. *Commun. ACM 68*, 7 (July 2025).
13. Li, P., Yang, J., Wierman, A., and Ren, S. Towards environmentally equitable AI via geographical load balancing. *e-Energy* (2024).
14. Maji, D., Shenoy, P., and Sitaraman, R.K. CarbonCast: Multi-day forecasting of grid carbon intensity. In *Proceedings of the 9th ACM Intern. Conf. on Systems for Energy-Efficient Buildings, Cities, and Transportation* (2022), 198–207.
15. Meta. Sustainability Report (2021); https://sustainability.fb.com/.
16. Mo, J. and Walrand, J. Fair end-to-end window-based congestion control. *IEEE/ACM Transactions on Networking 8*, 5 (2000), 556–567.
17. Murillo, J. et al. CDN-Shifter: Leveraging spatial workload shifting to decarbonize content delivery networks. In *Proceedings of the 2024 ACM Symp. on Cloud Computing*, 505–521.
18. OECD. Measuring the environmental impacts of artificial intelligence compute and applications: The AI footprint. *OECD Digital Economy Papers* (2022), 341; https://tinyurl.com/228ztx24
19. Patterson, D. et al. The carbon footprint of machine learning training will plateau, then shrink. *Computer 55*, 7 (2022), 18–28; 10.1109/MC.2022.3148714
20. Pessach, D. and Shmueli, E. A review on fairness in machine learning. *ACM Comput. Surv. 55*, 3, Article 51 (Feb. 2022), 44.
21. Radovanović, A. et al. Carbon-aware computing for datacenters. *IEEE Transactions on Power Systems 38*, 2 (2023), 1270–1280.
22. Srikant, R. and Ying, L. *Communication Networks: An Optimization, Control, and Stochastic Networks Perspective*. Cambridge University Press (2013)
23. UNESCO. Recommendation on the ethics of artificial intelligence. In *Policy Recommendation* (2022).
24. U.S. EPA. About the U.S. electricity system and its impact on the environment; https://tinyurl.com/y9ywl978.
25. Wu, C. et al. Sustainable AI: Environmental implications, challenges and opportunities. *In Proceedings of Machine Learning and Systems 4* (2022), 795–813.
26. Zhang, X. and Liu, M. *Fairness in Learning-Based Sequential Decision Algorithms: A Survey*. Springer International Publishing, Cham (2021) 525–555.

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